Zhexian Li

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Academic Positions

University of Michigan, Ann Arbor — Sep. 2026–present
Schmidt AI in Science Postdoctoral Fellow
Michigan Institute for Data and AI in Society
Mentors: Necmiye Ozay and Dimitra Panagou

Education

University of Southern California — Sep. 2021–Aug. 2026
Ph.D. in Civil Engineering
Advisor: Ketan Savla
Thesis: *Lifting-Based Control of Societal Systems*

University of Southern California — Sep. 2021–May 2026
M.S. in Electrical Engineering

Southeast University, Nanjing, China — Sep. 2016–June 2020
B.Eng. in Transportation Engineering

Visiting Positions

University of California, Santa Barbara — Sep. 2025–May 2026
Visiting Researcher
Department of Mechanical Engineering
Host: Bassam Bamieh

University of Minnesota, Minneapolis — Sep. 2019–May 2020
Visiting Researcher
Department of Civil, Environmental, and Geo-Engineering
Host: Michael W. Levin

Research Interests

Control, optimization, and learning of societal systems, including urban air mobility, road traffic networks, and water and energy networks, with guarantees on efficiency, resilience, and safety.

Journal Papers

[J8] **Z. Li**, F. P. J. de Barros, and K. Savla. Constrained bilinear optimal control of reactive evolution equations. *International Journal of Heat and Mass Transfer* 273, 129690 (2027). DOI: 10.1016/j.ijheatmasstransfer.2026.129690.

[J7] **Z. Li** and K. Savla. Output-feedback adaptive model predictive control for ramp metering: A set-membership approach. *Automatica* (2027). DOI: 10.1016/j.automatica.2026.113290.

[J6] **Z. Li**, A. S. Fokas, and K. Savla. A complex spatial frequency approach to optimal control of finite-extent linear evolution systems. *IEEE Transactions on Automatic Control* (2026). DOI: 10.1109/TAC.2026.3719216.

[J5] K. Han, W. Ji, Y. M. Nie, **Z. Li**, and S. Liu. Exploring the sensing power of mixed vehicle fleets. *Transportation Research Part B: Methodological* 190, 103066 (2024). DOI: 10.1016/j.trb.2024.103066.

[J4] **Z. Li**, M. W. Levin, X. Qu, and R. Stern. A network traffic model for the control of autonomous vehicles acting as moving bottlenecks. *IEEE Transactions on Intelligent Transportation Systems* 24.9 (2023), pp. 9004–9015. DOI: 10.1109/TITS.2023.3271187.

[J3] S. Wang, **Z. Li**, and M. W. Levin. Optimal policy for integrating autonomous vehicles into the auto market. *Transportation Research Part C: Emerging Technologies* 143, 103821 (2022). DOI: 10.1016/j.trc.2022.103821.

[J2] D. Kang, **Z. Li**, and M. W. Levin. Evasion planning for autonomous intersection control based on an optimized conflict point control formulation. *Journal of Transportation Safety & Security* 14.12 (2022), pp. 2074–2110. DOI: 10.1080/19439962.2021.1998939.

[J1] Y. Wang, **Z. Li**, Z. Ye, and Y. Zhang. Nighttime safety evaluation for signalized intersections at the signal cycle level based on rear-end conflict models considering lighting and traffic conditions. *Traffic Injury Prevention* 21.1 (2020), pp. 87–92. DOI: 10.1080/15389588.2019.1701670.

Conference Papers

[C7] **Z. Li** and K. Savla. FNO$^{\angle\theta}$: Extended Fourier neural operator for learning state and optimal control of distributed parameter systems. In: *2026 IEEE Conference on Decision and Control (CDC)*. 2026. https://arxiv.org/abs/2604.05187.

[C6] **Z. Li** and K. Savla. Adaptive model predictive control for traffic signal timing with unknown demand and parameters. In: *2025 American Control Conference (ACC)* (Denver, Colorado, July 2025). 2025, pp. 2210–2215. DOI: 10.23919/ACC63710.2025.11107474.

[C5] **Z. Li**, A. S. Fokas, and K. Savla. On linear quadratic regulator for the heat equation with general boundary conditions. In: *2024 IEEE 63rd Conference on Decision and Control (CDC)* (Milan, Italy, Dec. 2024). 2024, pp. 7594–7599. DOI: 10.1109/CDC56724.2024.10886399.

[C4] M. Pooladsanj, **Z. Li**, K. Savla, and P. Ioannou. Performance evaluation of ramp metering under different traffic measurement scenarios. In: *Transportation Research Board 103rd Annual Meeting* (Washington, D.C. Jan. 7–11, 2024). 2024. https://trid.trb.org/View/2334488.

[C3] **Z. Li** and K. Savla. Output-feedback model predictive control for ramp metering. In: *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)* (Bilbao, Spain, Sept. 2023). 2023, pp. 5259–5264. DOI: 10.1109/ITSC57777.2023.10421791.

[C2] **Z. Li**, M. W. Levin, R. Stern, and X. Qu. A network traffic model with controlled autonomous vehicles acting as moving bottlenecks. In: *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)* (Sept. 2020). 2020, pp. 1–6. DOI: 10.1109/ITSC45102.2020.9294607.

[C1] **Z. Li**, X. Zhu, X. Liu, and X. Qu. Model-based predictive variable speed limit control on multi-lane freeways with a line of connected automated vehicles. In: *2019 IEEE Intelligent Transportation Systems Conference (ITSC)* (Auckland, New Zealand, Oct. 2019). 2019, pp. 1989–1994. DOI: 10.1109/ITSC.2019.8917471.

Working Papers

Underlined names indicate mentored students.

[U2] **Z. Li**, N. Ozay, and D. Panagou. Online empty-vehicle routing with safe path-following rules for urban air mobility networks. Working paper. 2026.

[U1] **Z. Li**, J. A. Emhemed, and K. Savla. On disturbance-robust throughput for urban air mobility networks. Submitted to the 2027 American Control Conference (ACC). 2026.

Talks and Posters

2026	*Lifting is what you need for societal systems.* Seminar talk, Center for Control, Dynamical Systems, and Computation, University of California, Santa Barbara (April 8).
2026	*Lifting is what you need for societal systems.* Invited talk, Department of Decision and Control, KTH Royal Institute of Technology (March 23).
2026	*Lifting is what you need for societal systems.* Invited talk, School of Science and Engineering, Chinese University of Hong Kong, Shenzhen (January 29).
2025	*Predictive control for traffic networks with performance guarantees.* Posters at the Southern California Civil Engineering Symposium at USC, UC Irvine, and the Pacific Southwest Region University Transportation Center Annual Meeting at UC Berkeley.
2024	*A complex spatial frequency approach to optimal control of bounded spatially distributed systems.* Talk at the 44th Southern California Control Workshop.
2023	*Output-feedback ramp metering: model-based and model-free control.* Poster at the Southern California Civil Engineering Symposium at UCLA.

Honors and Awards

2026	USC Viterbi School of Engineering Best Research Assistant Award.
2023	Professor Teh-Fu Yen and Dr. Shia-Ping Siao Yen Fellowship.
2021	USC Viterbi/Graduate School Fellowship.
2020	Thesis Distinction in Jiangsu Province (top 0.5%).

Grant Writing

Physics-aware neural-operator reinforcement learning framework for fluid-flow energy-component design and operation 2026

PI: Ketan Savla; Co-PIs: Felipe de Barros and Birendra Jha

Assisted in conceptualization, data preparation, original drafting, and editing.
Target program: U.S. Department of Energy – Genesis Mission.

Predictive Control for Transport Systems with Performance Guarantees 2025
PI: Ketan Savla
Assisted in conceptualization, data preparation, original drafting, and editing.
Target program: National Science Foundation – Dynamics, Control and Systems Diagnostics.

Teaching Experience

University of Southern California Spring 2024 and 2025
Teaching Assistant, CE 330: Computer Methods in Civil Engineering
Student evaluation: 33.3% rated excellent; course score: 3.67/4.

University of Southern California Fall 2022 and 2023
Teaching Assistant, CE 471: Principles of Transportation Engineering
Student evaluation: 20% rated excellent; course score: 3.38/4.

Academic Service

Journal reviewer: ❑ Automatica ❑ Journal of Applied Mathematics and Physics (ZAMP) ❑ Transportation Research Part E: Logistics and Transportation Review.
Conference reviewer: ❑ IEEE Intelligent Transportation Systems Conference (ITSC) ❑ IEEE Conference on Decision and Control (CDC) ❑ American Control Conference (ACC) ❑ IFAC Modeling, Estimation and Control Conference (MECC) ❑ Transportation Research Board Annual Meeting (TRB).

Mentoring Experience

Voluntary Research Advisor
California Institute of Technology
Mentored senior undergraduate student J. A. Emhemed in the Department of Computing and Mathematical Sciences at Caltech.

Graduate Student Research Advisor 2024–present
University of Southern California
Mentored master's students Yajun Liu (Ph.D. Student at ASU), Zhengbang Liu (joined BYD), Xinyi Wu (Ph.D. Student at USC), and Zixiong Wang (joined HNTB).

Center for Undergraduate Research in Viterbi Engineering 2022–2024
University of Southern California
Mentored undergraduate research assistants Emma Silverstein, Kathryn Ruiz, and Aryan Shukla, and incoming Native American, Black, and Latinx engineering first-year undergraduate students at USC.

Industry Experience

China Blue Sky Environmental Technologies Sep. 2020–Jul. 2021
Software Engineer; Chengdu, China
Designed and implemented sensing vehicle routing optimization algorithms to improve coverage on a Python/Vissim simulation of a 50×50 traffic network in Chengdu, China.

Activities

2021–2023	Active member, SC Triathlon Club.
2019–2020	Active member, University of Minnesota Running Club.
2016–2019	College athlete, Southeast University Orienteering Team.